

Chapter 6

The Discrete Fourier Transform

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6.1 Sampling the Fourier Transform

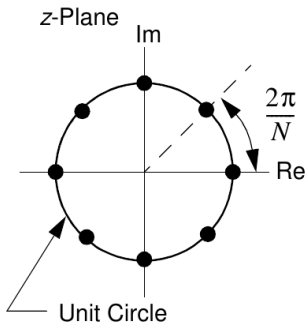
We have used the Fourier transform and the z-transform to analyze LTI systems. A third, computationally useful transform is the **discrete Fourier transform (DFT)**: a sequence of equally spaced samples of the DTFT of a finite-duration signal.

Assume the finite-length sequence $x[n]$ is zero everywhere except possibly for $n = 0, 1, \dots, N - 1$.

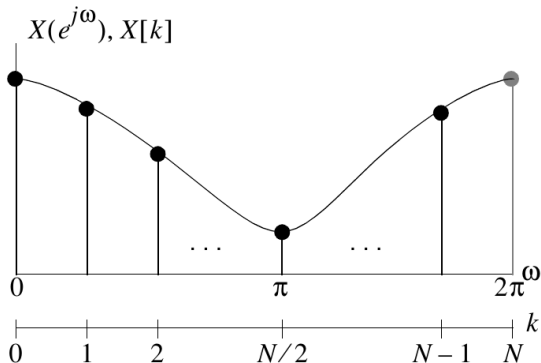
$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n},$$
$$\omega_k = \frac{2\pi k}{N}, \quad k = 0, 1, \dots, N - 1,$$
$$X(e^{j\omega_k}) = X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad W_N \triangleq e^{-j2\pi/N}.$$

- The DFT may be introduced through the discrete Fourier series (DFS), but here it is presented directly as frequency samples of the DTFT.

6.1 Sampling the Fourier Transform



(a) Sampling $X(z)$



(b) Sampling $X(e^{j\omega})$

6.2 The DFT–IDFT Pair

Analysis (DFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad k = 0, 1, \dots, N-1$$

Synthesis (IDFT)

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad n = 0, 1, \dots, N-1$$

- $W_N = e^{-j2\pi/N}$.
- The same algebraic pair appears in the DFT and DFS; the key distinction is the fixed-length interpretation of $x[n]$.

6.2 Verifying that the IDFT Returns $x[n]$

Substitute the DFT expression into the IDFT:

$$\begin{aligned}\frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} &= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{m=0}^{N-1} x[m] W_N^{km} \right] W_N^{-kn} \\ &= \sum_{m=0}^{N-1} x[m] \left[\frac{1}{N} \sum_{k=0}^{N-1} W_N^{k(m-n)} \right].\end{aligned}$$

The inner sum gives the discrete orthogonality relation

$$\sum_{k=0}^{N-1} W_N^{k(m-n)} = \begin{cases} N, & m = n, \\ 0, & m \neq n, \end{cases} \quad \text{so} \quad \frac{1}{N} \sum_{k=0}^{N-1} W_N^{k(m-n)} = \delta[m - n].$$

Therefore,

$$\frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} = x[n].$$

We use the shorthand notation $x[n] \xleftrightarrow{\text{DFT}} X[k]$.

6.2 Periodic Extensions Implied by the IDFT

When a finite-length signal is represented by a discrete Fourier series, periodic extension is implied. The notes use tildes to denote these periodic versions:

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} x[n + rN] = x[n \bmod N] = x[(n)_N],$$
$$\tilde{X}[k] = \sum_{r=-\infty}^{\infty} X[k + rN] = X[k \bmod N] = X[(k)_N].$$

- $\tilde{x}[n]$ denotes the finite sequence $x[n]$ together with all of its periodic repetitions.
- $\tilde{X}[k]$ plays the same role in the transform domain.
- This keeps the time-domain and frequency-domain descriptions dual to one another.

6.2 Matrix View of the DFT

Let

$$x_N = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}, \quad X_N = \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix},$$

and define the DFT matrix

$$W_N = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N & W_N^2 & \cdots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)(N-1)} \end{bmatrix}.$$

Then

$$X_N = W_N x_N, \quad x_N = W_N^{-1} X_N = \frac{1}{N} W_N^* X_N.$$

6.2 Matrix View of the DFT

$$W_N = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N & W_N^2 & \cdots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)(N-1)} \end{bmatrix}.$$

Notice that

$$W_N^{-1} = \frac{1}{N} W_N^*, \quad W_N W_N^* = N I_N.$$

so the DFT matrix is orthogonal up to the scale factor N .

Example 6.1: A Pulse Sequence

Consider the N -point DFT of

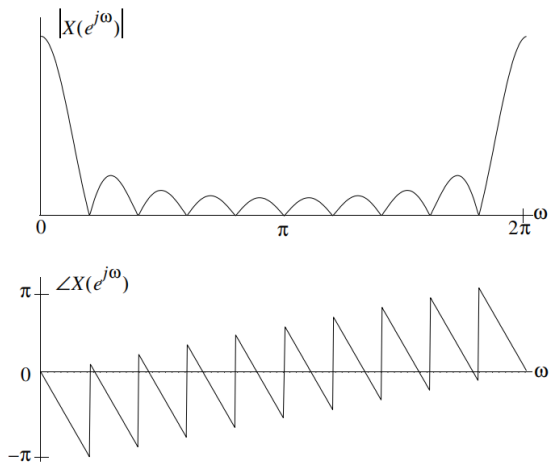
$$x[n] = \begin{cases} 1, & 0 \leq n \leq L-1, \\ 0, & \text{otherwise,} \end{cases} \quad N \geq L.$$

Its DTFT is

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{L-1} e^{-j\omega n} = \frac{1 - e^{-j\omega L}}{1 - e^{-j\omega}} \\ &= \frac{\sin(\omega L/2)}{\sin(\omega/2)} e^{-j\omega(L-1)/2}. \end{aligned}$$

- The magnitude is the familiar sampled-sinc envelope.
- The phase is linear in ω except at the zeros of the magnitude.

Example 6.1: A Pulse Sequence



Magnitude and phase of $X(e^{j\omega})$ for $L = 10$

Example 6.1: Frequency Sampling with $N = L$

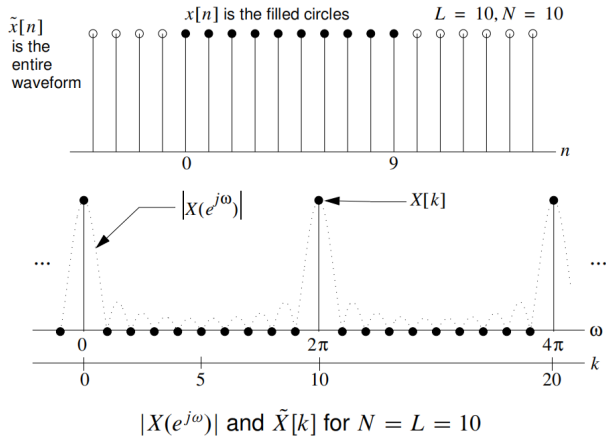
Sampling the DTFT at $\omega_k = 2\pi k/N$ gives

$$X[k] = \frac{\sin(\pi kL/N)}{\sin(\pi k/N)} e^{-j\pi k(L-1)/N}, \quad k = 0, 1, \dots, N-1.$$

If $N = L$, then

$$X[k] = \begin{cases} L, & k = 0, \\ 0, & k = 1, 2, \dots, L-1. \end{cases}$$

Example 6.1: Frequency Sampling with $N = L$



The DFT sees the periodically extended sequence rather than an isolated pulse. Clearly, using too few frequency samples causes a loss of spectral detail.

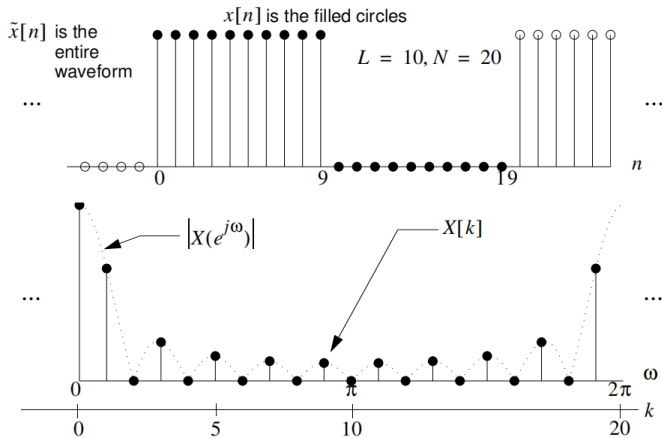
Example 6.1: Zero Padding to Increase Frequency Samples

By appending $N - L$ zeros, we obtain a denser sampling of the *same* underlying DTFT. If $L = 10$ and the transform length is increased to $N = 20$,

- the periodic extension becomes ten ones followed by ten zeros, and then repeats;
- the interval $0 \leq \omega \leq 2\pi$ now contains twice as many DFT samples.

Example 6.1: Zero Padding to Increase Frequency Samples

By appending $N - L$ zeros, we obtain a denser sampling of the *same* underlying DTFT. If $L = 10$ and the transform length is increased to $N = 20$.

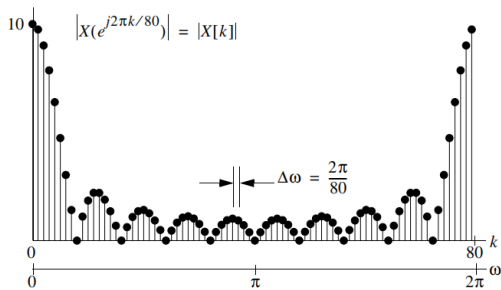


$|X(e^{j\omega})|$ and $\tilde{X}[k]$ for $L = 10, N = 20$

Example 6.1: Zero Padding Further to $N = 80$

We push the same example to a much longer DFT length.

- The signal still contains only ten nonzero samples.
- The extra zero padding does *not* change the DTFT.
- It only gives a finer set of DFT samples, so the DFT better approximates the true continuous curve.



Magnitude and phase of $X[k]$ for $L = 10$ and $N = 80$

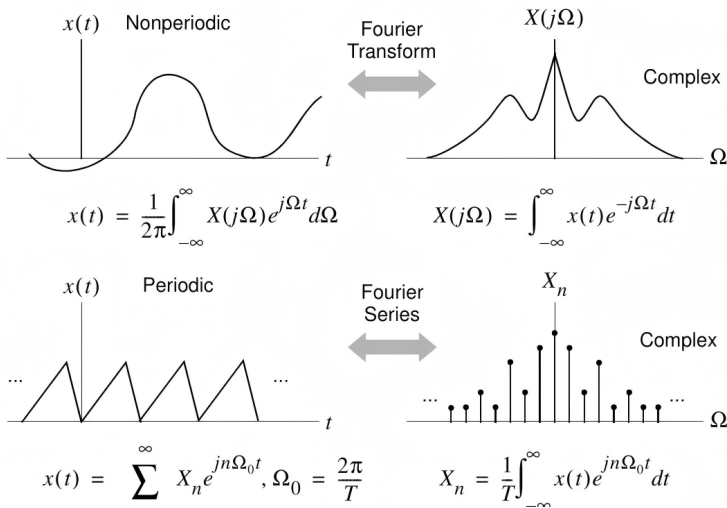
6.3 Summary of Fourier Representations: Continuous Time

Followings summarize how Fourier representations relate periodicity and spectral description.

For continuous-time signals,

- Continuous-time, nonperiodic signals use the Fourier transform.
- Continuous-time, periodic signals use the Fourier series.
- In both cases the coefficients are, in general, complex-valued.

6.3 Summary of Fourier Representations: Continuous Time



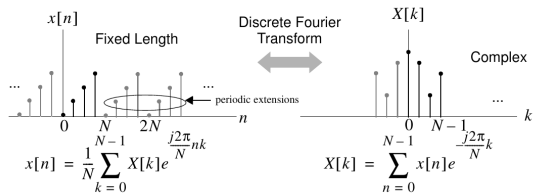
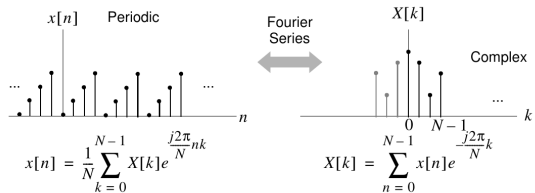
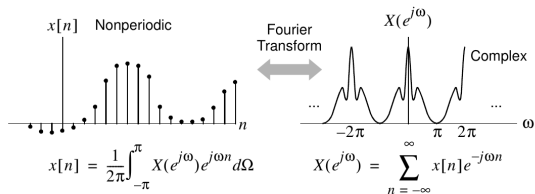
Continuous-time Fourier transform and Fourier series relationships.

6.3 Summary of Fourier Representations: Discrete Time

For discrete-time signals,

- the DTFT for nonperiodic sequences;
- the discrete-time Fourier series for periodic sequences;
- the DFT for fixed-length sequences with implicit periodic extension.

6.3 Summary of Fourier Representations: Discrete Time



6.4 Linearity

If $x_1[n]$ and $x_2[n]$ have lengths N_1 and N_2 , then after zero padding to any common length $N \geq \max\{N_1, N_2\}$,

$$ax_1[n] + bx_2[n] \xleftrightarrow{\text{DFT}} aX_1[k] + bX_2[k].$$

6.4 Circular Sequence Shift

From the DTFT we know that

$$x[n - m] \xleftrightarrow{\mathcal{F}} e^{-j\omega m} X(e^{j\omega}).$$

For the DFT, the corresponding property becomes

$$x[n] \xleftrightarrow{\text{DFT}} X[k], \quad x_1[n] \xleftrightarrow{\text{DFT}} W_N^{mk} X[k],$$

where

$$x_1[n] = x[((n - m))_N].$$

Because the DFT assumes periodic extension, the time-domain shift becomes a **circular** shift rather than an ordinary shift.

6.4 Circular Sequence Shift: Derivation

Using the IDFT,

$$\begin{aligned}x_1[n] &= \frac{1}{N} \sum_{k=0}^{N-1} W_N^{mk} X[k] W_N^{-nk} \\&= \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-(n-m)k} \\&= x[\langle (n-m) \rangle_N], \quad 0 \leq n \leq N-1.\end{aligned}$$

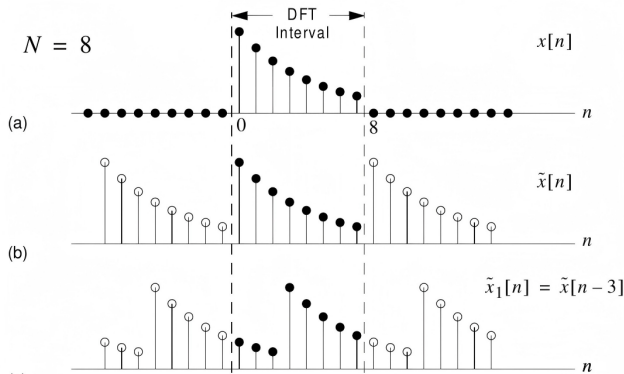
So the circular-shift property is

$$x[\langle (n-m) \rangle_N] \xleftrightarrow{\text{DFT}} W_N^{km} X[k].$$

The key point is that the time shift is taken modulo N , so samples that move past one end of the DFT interval re-enter from the other end.

6.4 Circular Sequence Shift: Graphical Interpretation

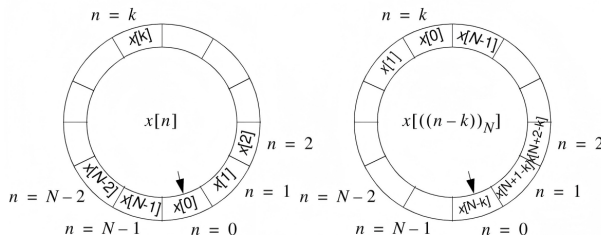
We can illustrate the result by shifting the periodically extended sequence by $m = 3$. Over a single DFT interval, the rightmost samples wrap around and reappear at the left edge.



A circular shift by $m = 3$ viewed over one DFT interval.

6.4 Circular Sequence Shift Viewed on a Cylinder

The figure give a nice geometric interpretation: place the samples of $x[n]$ around the circumference of a cylinder. Shifting by m simply rotates the sample locations around the cylinder, so when the sequence is "unwrapped" again, the shifted samples re-enter from the opposite side.



Circular shift viewed on a cylinder

The periodic indexing in the DFT makes the shift circular, not linear.

$$x[((n - m))_N] \xleftrightarrow{\text{DFT}} W_N^{km} X[k]$$

6.4 Duality

Assume

$$x[n] \xleftrightarrow{\text{DFT}} X[k].$$

Then the DFT of the sequence $X[n]$ is

$$\begin{aligned}\text{DFT}\{X[n]\} &= \sum_{n=0}^{N-1} X[n] W_N^{kn} \\ &= N x[\left((-k)\right)_N] \\ &= N x[N - k], \quad 0 \leq k \leq N - 1.\end{aligned}$$

Thus the DFT duality relation is

$$X[n] \xleftrightarrow{\text{DFT}} N x[\left((-k)\right)_N] = N x[N - k].$$

This is the DFT analogue of the duality relationship known from the continuous Fourier transform.

6.4 Symmetry Properties: Two Fundamental Relations

Two identities drive the symmetry results:

$$x^*[n] \xleftrightarrow{\text{DFT}} X^*[((-k))_N], \quad x^*[((-n))_N] \xleftrightarrow{\text{DFT}} X^*[k].$$

We can prove them directly from the DFT and IDFT definitions.

$$\begin{aligned} \text{DFT}\{x^*[n]\} &= \sum_{n=0}^{N-1} x^*[n] W_N^{kn} = \left[\sum_{n=0}^{N-1} x[n] W_N^{-kn} \right]^* \\ &= X^*[((-k))_N], \end{aligned}$$

and similarly,

$$\text{IDFT}\{X^*[k]\} = \frac{1}{N} \sum_{k=0}^{N-1} X^*[k] W_N^{-kn} = x^*[((-n))_N].$$

These control the rest of the symmetry discussion.

6.4 Symmetry Properties: Conjugate Parts and Real Sequences

Define the periodic conjugate-symmetric and conjugate-antisymmetric parts of $x[n]$ by

$$\begin{aligned}x_{ep}[n] &= \frac{1}{2} \left(x[n] + x^*[N - n] \right), \\x_{op}[n] &= \frac{1}{2} \left(x[n] - x^*[N - n] \right), \quad 0 \leq n \leq N - 1.\end{aligned}$$

Then

$$\begin{aligned}\Re\{x[n]\} &\xleftrightarrow{\text{DFT}} X_{ep}[k] = \frac{1}{2} \left(X[k] + X^*[N - k] \right), \\j \Im\{x[n]\} &\xleftrightarrow{\text{DFT}} X_{op}[k] = \frac{1}{2} \left(X[k] - X^*[N - k] \right), \\x_{ep}[n] &\xleftrightarrow{\text{DFT}} \Re\{X[k]\}, \\x_{op}[n] &\xleftrightarrow{\text{DFT}} j \Im\{X[k]\}.\end{aligned}$$

For a real sequence $x[n]$,

$$X[k] = X^*[N - k], \quad |X[k]| = |X[N - k]|, \quad \angle X[k] = -\angle X[N - k].$$

So for real-valued data the DFT is unique over only the first $N/2$ frequency samples.

6.4 Circular Convolution

Suppose two N -point sequences $x_1[n]$ and $x_2[n]$ have DFTs $X_1[k]$ and $X_2[k]$. Let

$$X_3[k] = X_1[k]X_2[k].$$

Taking the IDFT,

$$\begin{aligned}x_3[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X_1[k]X_2[k]W_N^{-kn} \\&= \sum_{m=0}^{N-1} x_1[m] x_2[\langle n - m \rangle_N], \quad 0 \leq n \leq N - 1.\end{aligned}$$

The right-hand side is the **N -point circular convolution** of $x_1[n]$ and $x_2[n]$.

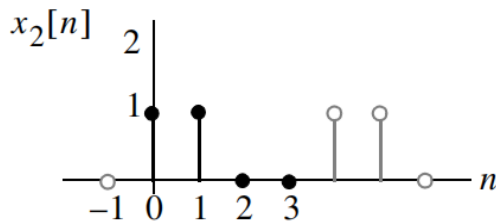
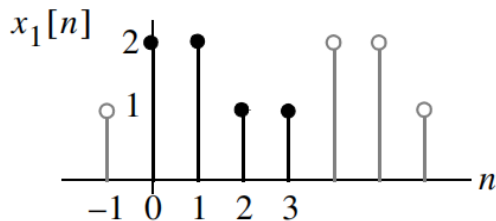
$$x_3[n] = x_1[n] \circledast_N x_2[n] = \sum_{m=0}^{N-1} x_1[m]x_2[\langle n - m \rangle_N]$$

Like linear convolution, circular convolution is commutative.

Example 6.2: 4-Point Circular Convolution

Compute the circular convolution of a 4-point example with

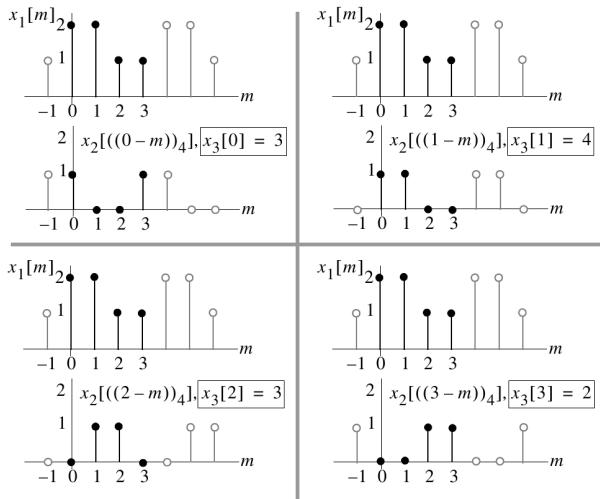
$$x_1[n] = [2, 2, 1, 1], \quad x_2[n] = [1, 1, 0, 0].$$



Input sequences used in the 4-point circular-convolution example.

Example 6.2: Graphical Evaluation

For each $n = 0, 1, 2, 3$, the periodically extended sequence $x_2[(n - m)]_4$ is rotated against $x_1[m]$ and then summed. That graphical construction gives the circular-convolution samples directly.



Example 6.2: Transform Values $X_1[k]$ and $X_2[k]$

The corresponding DFT values are

$$\begin{aligned}X_1[k] &= 2 + 2e^{-j\pi k/2} + e^{-j\pi k} + e^{-j3\pi k/2}, \\X_1[0] &= 6, \quad X_1[1] = 1 - j, \quad X_1[2] = 0, \quad X_1[3] = 1 + j, \\X_2[k] &= 1 + e^{-j\pi k/2}, \\X_2[0] &= 2, \quad X_2[1] = 1 - j, \quad X_2[2] = 0, \quad X_2[3] = 1 + j.\end{aligned}$$

These values match the graphical construction and set up the final IDFT step.

Example 6.2: Product Spectrum and IDFT

Form the pointwise product:

$$X_3[0] = 12, \quad X_3[1] = -2j, \quad X_3[2] = 0, \quad X_3[3] = 2j.$$

Hence

$$x_3[n] = \frac{1}{4} \left(12 - 2je^{j\pi n/2} + 2je^{j3\pi n/2} \right),$$

yielding

$$x_3[0] = 3, \quad x_3[1] = 4, \quad x_3[2] = 3, \quad x_3[3] = 2,$$

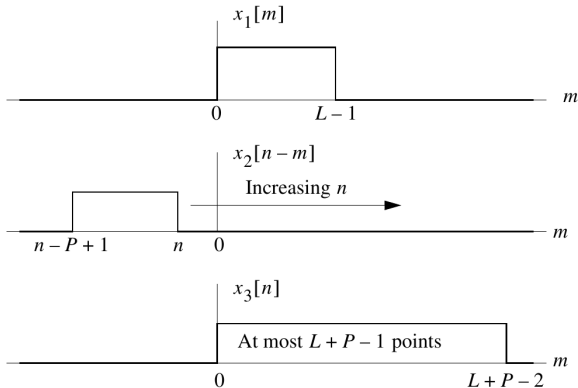
which is the expected result.

6.4 Time Aliasing in Circular Convolution

For finite-length sequences $x_1[n]$, $0 \leq n \leq L - 1$, and $x_2[n]$, $0 \leq n \leq P - 1$, the linear convolution is

$$x_3[n] = \sum_{m=-\infty}^{\infty} x_1[m]x_2[n - m].$$

Because both sequences are finite, $x_3[n]$ can have at most $L + P - 1$ nonzero values.



6.4 Circular Convolution Model

The corresponding N -point circular convolution is

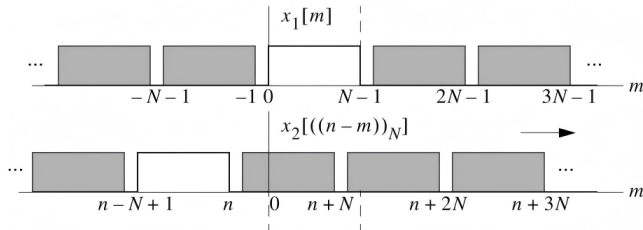
$$x_{3p}[n] = \sum_{m=0}^{N-1} x_1[m]x_2[((n - m))_N],$$

where the transform length N is chosen by the designer.

The key question is whether this periodic convolution matches the ordinary linear convolution over the interval of interest. In general it does not; the difference shows up as time aliasing unless sufficient zero padding is used.

6.4 Time Aliasing as a Periodic Sum of Linear Convolutions

Notice that circular convolution produces a *periodic* sequence. The periodic extensions of $x_1[m]$ and $x_2[(n-m)]_N$ overlap repeatedly, which leads directly to time aliasing.



Periodic extensions inside the circular-convolution sum.

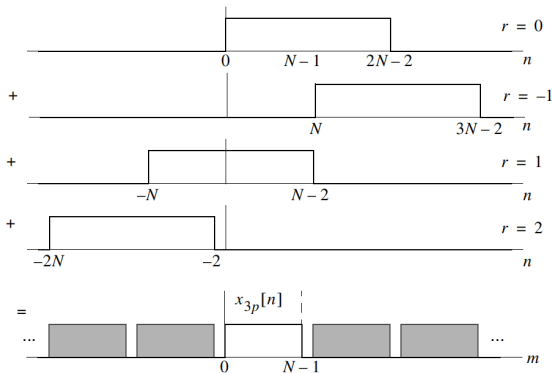
That observation leads to the key relation

$$x_{3p}[n] = \sum_{r=-\infty}^{\infty} x_3[n + rN].$$

So the circular-convolution result is a periodic superposition of shifted copies of the linear-convolution result.

6.4 Superposition of Shifted Linear Convolutions

Unpack the previous formula geometrically by drawing the translates of $x_3[n]$ that contribute to $x_{3p}[n]$.



The formation of $x_{3p}[n]$ as a sum of translates of $x_3[n]$

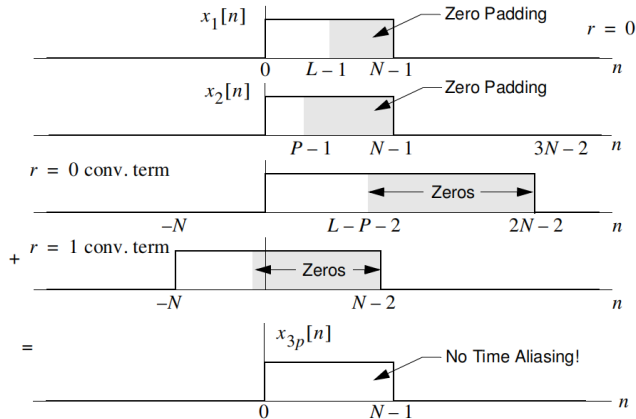
On the interval $0 \leq n \leq N-1$, the dominant aliasing contribution comes from the $r=1$ translate, namely $x_3[n+N]$. In general, circular and linear convolution are therefore *not* equivalent.

6.4 Avoiding Time Aliasing

If we choose the transform length large enough,

$$N \geq L + P - 1,$$

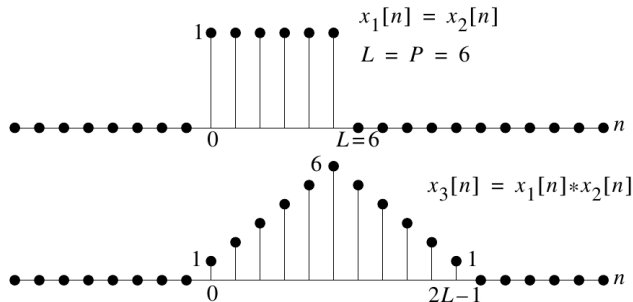
then the shifted replicas no longer overlap the interval of interest, and the circular convolution reproduces the linear convolution over one N -sample block.



Example 6.3: Two Six-Point Sequences

Compare linear convolution and N -point circular convolution for

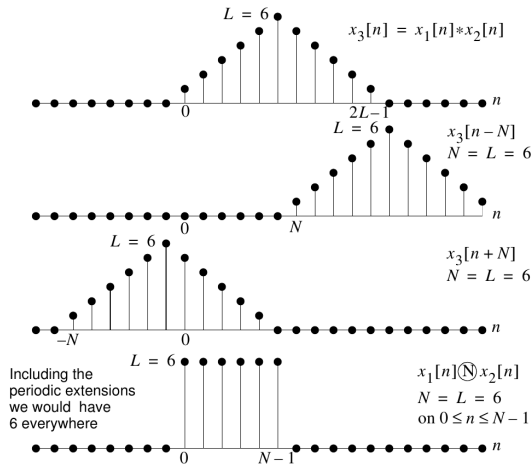
$$x_1[n] = x_2[n] = \begin{cases} 1, & 0 \leq n \leq 5, \\ 0, & \text{otherwise,} \end{cases} \quad L = P = 6.$$



The linear convolution of two length-6 rectangles is the familiar triangular sequence.

Example 6.3: Circular Convolution with $N = 6$

If the circular-convolution length is chosen as $N = 6$, then the periodic replicas overlap badly. Over the interval $0 \leq n \leq N - 1$, the result is obtained by adding $x_3[n]$ and $x_3[n + N]$.



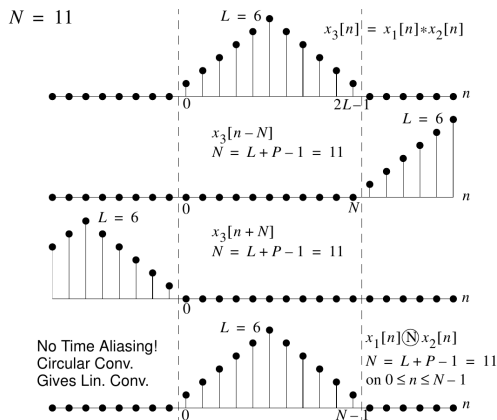
Severe time aliasing when the circular-convolution length is only $N = 6$.

Example 6.3: Choosing $N = 11$

To avoid time aliasing, it requires

$$N \geq L + P - 1 = 11.$$

With this choice, the circular convolution matches the linear convolution on $0 \leq n \leq N - 1$.



No time aliasing when the transform length is large enough.

6.4 Multiplication Theorem and Parseval's Theorem

Multiplication theorem is the dual of the circular-convolution theorem.

$$x_1[n]x_2[n] \xleftrightarrow{\text{DFT}} \frac{1}{N} \sum_{m=0}^{N-1} X_1[m]X_2[(k-m)_N]$$

Parseval's theorem for the DFT is

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2.$$

6.4 Summary of Properties

Finite-length sequence relation	N -point DFT relation
$ax_1[n] + bx_2[n]$	$aX_1[k] + bX_2[k]$
$X[n]$	$Nx[(-k))_N]$
$x[((n-m))_N]$	$W_N^{km}X[k]$
$W_N^{\ell n}x[n]$	$X[((k-\ell))_N]$
$\sum_{m=0}^{N-1} x_1[m]x_2[((n-m))_N]$	$X_1[k]X_2[k]$
$x_1[n]x_2[n]$	$\frac{1}{N} \sum_{\ell=0}^{N-1} X_1[\ell]X_2[((k-\ell))_N]$
$x^*[n]$	$X^*[(-k))_N]$
$x^*[((-n))_N]$	$X^*[k]$
For real $x[n]$	$X[k] = X^*[(-k))_N], X[k] = X[(-k))_N] ,$ $\angle X[k] = -\angle X[(-k))_N]$

6.5 Implementation of LTI Systems using the DFT

Suppose an input sequence $x[n]$ has length L and an impulse response $h[n]$ has length P . To compute the linear convolution using the DFT, choose a transform length satisfying

$$N \geq L + P - 1.$$

Then:

- append zeros to $x[n]$ and $h[n]$ so both sequences have length N ;
- compute their N -point DFTs, $X[k]$ and $H[k]$;
- multiply to form $Y[k] = X[k]H[k]$;
- compute the IDFT of $Y[k]$ to recover $y[n] = x[n] * h[n]$.

Notice that the zero padding is what prevents time aliasing and makes the circular-convolution machinery perform true linear convolution.

6.5 Practical Transform-Domain Convolution Steps

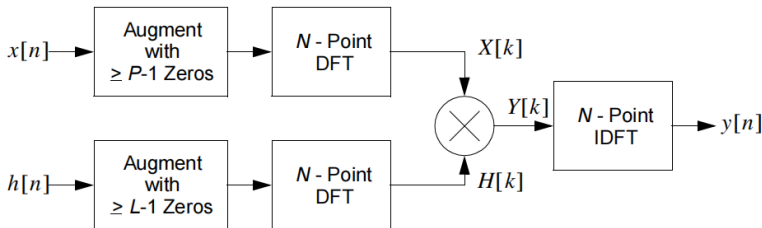
We write the zero-padded sequences as

$$\begin{aligned}\{x[n]\} &\rightarrow \{x[0], x[1], \dots, x[L-1], 0, \dots, 0\}, \\ \{h[n]\} &\rightarrow \{h[0], h[1], \dots, h[P-1], 0, \dots, 0\},\end{aligned}$$

with both augmented sequences having total length $N \geq L + P - 1$.

- First compute the N -point DFT of each augmented sequence.
- Next multiply the spectra pointwise to obtain $Y[k]$.
- Finally apply the N -point IDFT to obtain the linear-convolution result $y[n]$.

6.5 Practical Transform-Domain Convolution Steps



Using the DFT to convolve two finite-length sequences